

Statistics

Mean, median and mode of grouped data, with direct, assumed-mean and step-deviation methods, formulae and worked examples.

Introduction

In Class 9 you found the mean, median and mode of ungrouped data. In Class 10, data is often grouped into class intervals with a frequency for each class. Because the exact observations inside a class are not available, the class mark is used as a representative value while calculating the mean.

Class mark

For a class interval, class mark $x_i = (\text{upper limit} + \text{lower limit}) / 2$. For example, the class mark of 20, 30 is 25.

Mean of Grouped Data

There are three methods for finding the mean of grouped data. All three give the same answer, you can use whichever makes the arithmetic easiest.

1. Direct method

Mean, Direct method

$$\text{Mean } (\bar{x}) = \frac{\sum f_i x_i}{\sum f_i}$$

where f_i is the frequency of class i and x_i is its class mark. $\sum f_i = n$ is the total number of observations.

This is straightforward but involves multiplying large class marks, so it can be heavy when the numbers are big.

2. Assumed mean method

Pick any class mark as the **assumed mean** a (choosing the middle class usually keeps numbers small). Let $d_i = x_i - a$ be the deviation of each class mark from a .

Mean, Assumed mean method

$$\bar{x} = a + \frac{\sum f_i d_i}{\sum f_i}$$

Because d_i values are smaller than x_i values, this often makes calculations easier.

3. Step-deviation method

If all class intervals have the same width h , we can simplify further. Let $u_i = (x_i - a)/h$. Since h divides evenly, u_i values are small integers.

Mean, Step-deviation method

$$\bar{x} = a + \left(\frac{\sum f_i u_i}{\sum f_i} \right) \times h$$

where $u_i = (x_i - a)/h$ and h is the class size. This is the shortest method when the class size is constant, and is the one most often expected in the exam for grouped data.

Worked Example 1, Mean by step-deviation

Find the mean of the following data:

Classes: 0, 10 ($f=5$), 10, 20 ($f=8$), 20, 30 ($f=12$), 30, 40 ($f=10$), 40, 50 ($f=5$).

Class marks x_i : 5, 15, 25, 35, 45. Class size $h = 10$. Choose assumed mean $a = 25$ (middle class).

Compute $u_i = (x_i - 25)/10$: -2, -1, 0, 1, 2.

$f_i u_i$: $5 \times (-2) = -10$, $8 \times (-1) = -8$, $12 \times 0 = 0$, $10 \times 1 = 10$, $5 \times 2 = 10$. Sum $\sum f_i u_i = 2$.

Total $\sum f_i = 40$.

$$\bar{x} = 25 + (2/40) \times 10 = 25 + 0.5 = \mathbf{25.5}$$

You can verify with the direct method: $\sum f_i x_i = 25 + 120 + 300 + 350 + 225 = 1020$, and $1020/40 = 25.5$.

All three methods give the same mean. In practice, the step-deviation method saves the most time, especially when class marks are large.

Mode of Grouped Data

The **mode** is the value that occurs most often. For grouped data, we first find the **modal class** (the class with the highest frequency) and then use a formula that estimates where the peak lies inside that class.

Mode of grouped data

$$\text{Mode} = l + (f_1 - f_0) / (2f_1 - f_0 - f_2) \times h$$

where l = lower limit of the modal class, h = class size, f_1 = frequency of the modal class, f_0 = frequency of the class before the modal class, f_2 = frequency of the class after the modal class.

If the highest frequency occurs in the first or last class, one of f_0 or f_2 is taken as 0.

Worked Example 2, Mode

Find the mode of the data:

Classes: 0, 10 ($f=4$), 10, 20 ($f=8$), 20, 30 ($f=15$), 30, 40 ($f=10$), 40, 50 ($f=6$).

The highest frequency is 15, so the modal class is 20, 30. Here $l = 20$, $h = 10$, $f_1 = 15$, $f_0 = 8$, $f_2 = 10$.

$$\text{Mode} = 20 + (15 - 8) / (30 - 8 - 10) \times 10 = 20 + 7/12 \times 10 = 20 + 5.83 = \mathbf{25.83} \text{ (approx.)}$$

Remember: the grouped-data formulas in this chapter assume the classes are continuous, that is, the upper limit of one class equals the lower limit of the next. If the data is given in non-continuous form (for example, 0, 9, 10, 19), convert it to continuous form (0, 9.5, 9.5, 19.5) before applying these formulas, but such cases are rare in Class 10.

Median of Grouped Data

The **median** is the middle value when all observations are arranged in order. For grouped data, we use cumulative frequencies to find the class that contains the middle observation.

Median of grouped data

$$\text{Median} = l + (n/2 - cf) / f \times h$$

where l = lower limit of the median class, $n = \sum f_i$ (total observations), cf = cumulative frequency of the class before the median class, f = frequency of the median class, h = class size. The median class is the class whose cumulative frequency first reaches or exceeds $n/2$.

Worked Example 3, Median

Find the median of the data:

Classes: 0, 10 ($f=5$), 10, 20 ($f=7$), 20, 30 ($f=10$), 30, 40 ($f=8$), 40, 50 ($f=4$). $n = 34$.

Cumulative frequencies: 5, 12, 22, 30, 34. Need $n/2 = 17$.

The cumulative frequency first reaches 17 in the class 20, 30 ($cf = 22$). So median class is 20, 30. Here $l = 20$, cf of previous class = 12, $f = 10$, $h = 10$.

$$\text{Median} = 20 + (17 - 12) / 10 \times 10 = 20 + 5 = \mathbf{25}$$

If n is odd, the median is still given by the same formula, $n/2$ may be a .5 value, but the same median class rule applies. When every class has equal width, the formula works cleanly.

A useful relationship that sometimes appears for reference (not required to derive in the exam) is:

$$3 \text{ Median} = \text{Mode} + 2 \text{ Mean}$$

This is an approximate empirical relationship. It is good for a quick check, but do not use it to calculate one average from the others in a formal question unless the question itself asks you to verify it.

Summary

- **Class mark** $x_i = (\text{upper limit} + \text{lower limit})/2$ is used as the representative value for grouped data.
- **Mean, Direct:** $\bar{x} = \Sigma f_i x_i / \Sigma f_i$.
- **Mean, Assumed mean:** $\bar{x} = a + \Sigma f_i d_i / \Sigma f_i$, where $d_i = x_i - a$.
- **Mean, Step-deviation:** $\bar{x} = a + (\Sigma f_i u_i / \Sigma f_i) \times h$, where $u_i = (x_i - a)/h$. Best when class size h is constant.
- **Mode:** $\text{Mode} = l + (f_1 - f_0)/(2f_1 - f_0 - f_2) \times h$. Modal class = class with highest frequency.
- **Median:** $\text{Median} = l + (n/2 - cf)/f \times h$. Median class = first class with cumulative frequency $\geq n/2$.
- Empirical check (approximate): $3 \text{ Median} = \text{Mode} + 2 \text{ Mean}$.

Practice Questions

1. The following table gives the number of books read by 40 students in a month. Find the mean using the step-deviation method: 0, 5 ($f=4$), 5, 10 ($f=8$), 10, 15 ($f=12$), 15, 20 ($f=10$), 20, 25 ($f=6$).

2. Find the mode of: 0, 10 ($f=6$), 10, 20 ($f=10$), 20, 30 ($f=16$), 30, 40 ($f=12$), 40, 50 ($f=8$).

3. Find the median of: 0-20 ($f=5$), 20-40 ($f=10$), 40-60 ($f=15$), 60-80 ($f=8$), 80-100 ($f=6$).

4. In a grouped data, the mean is 25.5 and the mode is 25.83. Estimate the median using the empirical relationship.

5. A class has marks: 10, 20 ($f=3$), 20, 30 ($f=7$), 30, 40 ($f=9$), 40, 50 ($f=6$), 50, 60 ($f=5$). Find the mean by the assumed mean method taking $a = 35$, and verify by the direct method.